

DATA ANALYTICS – PROJECT 1

Data Cleaning & Preparation

Intern Details

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Organization: Decode Labs

Project: Project 1 – Data Cleaning & Preparation

Objective

The objective of this project is to clean and prepare a raw dataset for further analysis by identifying missing values, checking duplicate records, and validating data formats.

Data cleaning ensures that the dataset is accurate, consistent, and reliable for decision-making.

Dataset Overview

The provided dataset contains transactional data with multiple attributes such as order details, customer information, product information, and coupon usage. Before performing any analysis, the dataset was examined for data quality issues.

Tasks Performed

1. Missing Value Identification

The dataset was inspected to identify missing or null values across all columns.

Observation:

- Missing values were found in the **Coupon Code** column.

Action Taken:

- Missing Coupon Code entries were handled appropriately to maintain data consistency.

Screenshot:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	OrderID	Date	Customer	Product	Quantity	UnitPrice	ShippingAmount	PaymentMethod	OrderStatus	TrackingNumber	ItemsInOrder	CouponCode	ReferralSource	TotalPrice			
10	ORD200008	#####	C26817	Phone	2	134.28	904	Main	Gift Card	Cancelled	TRK610421	2	No Coupon	Email			268.56
17	ORD200015	#####	C39416	Printer	1	473.96	942	Main	Cash	Delivered	TRK549304	3	No Coupon	Google			473.96
19	ORD200017	#####	C60672	Tablet	1	423.4	382	Main	Debit Card	Shipped	TRK119721	3	No Coupon	Referral			423.4
23	ORD200021	#####	C99023	Monitor	4	342.95	569	Main	Cash	Delivered	TRK526461	6	No Coupon	Google			1371.8
28	ORD200026	#####	C21944	Laptop	5	59.55	296	Main	Online	Delivered	TRK802761	6	No Coupon	Referral			297.75
30	ORD200028	#####	C59823	Phone	4	160.07	568	Main	Online	Pending	TRK658951	5	No Coupon	Email			640.28
33	ORD200031	#####	C26445	Chair	3	500.08	996	Main	Debit Card	Returned	TRK312291	6	No Coupon	Facebook			1500.24
34	ORD200032	#####	C12388	Tablet	5	536.72	830	Main	Cash	Delivered	TRK992611	10	No Coupon	Email			2683.6
38	ORD200036	#####	C59337	Laptop	2	534.43	465	Main	Debit Card	Pending	TRK509611	5	No Coupon	Facebook			1068.86
14	ORD200042	#####	C28712	Printer	2	151.66	980	Main	Cash	Cancelled	TRK407721	6	No Coupon	Instagram			303.32
15	ORD200043	#####	C60572	Chair	3	499.21	871	Main	Online	Cancelled	TRK203511	8	No Coupon	Instagram			1497.63
50	ORD200048	#####	C12408	Monitor	2	80.77	464	Main	Debit Card	Shipped	TRK389381	4	No Coupon	Google			161.54
57	ORD200055	#####	C95984	Printer	3	344.4	565	Main	Online	Cancelled	TRK674341	3	No Coupon	Facebook			1033.2
31	ORD200059	#####	C97603	Chair	4	593.27	811	Main	Online	Shipped	TRK470551	5	No Coupon	Facebook			2373.08
57	ORD200065	#####	C97735	Monitor	5	81.7	385	Main	Credit Card	Delivered	TRK137431	6	No Coupon	Instagram			408.5
59	ORD200067	#####	C25587	Printer	4	12.1	440	Main	Debit Card	Cancelled	TRK551051	8	No Coupon	Google			48.4
71	ORD200069	#####	C43338	Monitor	3	478.65	552	Main	Online	Shipped	TRK352411	4	No Coupon	Referral			1435.95
73	ORD200071	#####	C37454	Monitor	3	434.45	923	Main	Credit Card	Pending	TRK282521	4	No Coupon	Referral			1303.35
74	ORD200072	#####	C19451	Tablet	4	589.44	942	Main	Gift Card	Returned	TRK877871	5	No Coupon	Facebook			2357.76
75	ORD200073	#####	C17269	Monitor	4	433.45	129	Main	Gift Card	Cancelled	TRK796081	4	No Coupon	Facebook			1733.8
76	ORD200074	#####	C25382	Phone	5	90.87	622	Main	Cash	Pending	TRK646911	5	No Coupon	Instagram			454.35
79	ORD200077	#####	C50121	Phone	1	583.18	348	Main	Credit Card	Returned	TRK798101	5	No Coupon	Google			583.18
31	ORD200079	#####	C76700	Tablet	2	311.91	103	Main	Online	Shipped	TRK130361	2	No Coupon	Facebook			623.82
32	ORD200080	#####	C29041	Phone	3	589.56	591	Main	Online	Returned	TRK288401	6	No Coupon	Facebook			1768.68
35	ORD200083	#####	C15681	Laptop	4	540.41	989	Main	Credit Card	Cancelled	TRK751121	6	No Coupon	Referral			2161.64
36	ORD200088	#####	C75439	Desk	3	114.81	318	Main	Gift Card	Cancelled	TRK198311	7	No Coupon	Google			344.43
37	ORD200093	#####	C67363	Desk	4	473.01	833	Main	Credit Card	Returned	TRK669361	7	No Coupon	Referral			1734.04

2. Duplicate Record Verification

The dataset was checked for duplicate records and duplicate Order IDs.

Observation:

- No duplicate records were found.
- No duplicate Order IDs were identified.

Result:

- Dataset uniqueness was maintained.

Screenshot:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	OrderID	Date	Customer	Product	Quantity	UnitPrice	ShippingAmount	PaymentMethod	OrderStatus	TrackingNumber	ItemsInOrder	CouponCode	ReferralSource	TotalPrice				
2	ORD200000	#####	C72649	Monitor	5	570.62	928	Main	Debit Card	Shipped	TRK379471	7	SAVE10	Instagram				2853.1
3	ORD200001	#####	C75739	Phone	2	151.35	823	Main	Online	Shipped	TRK911861	3	SAVE10	Referral				302.7
4	ORD200002	#####	C81728	Tablet	5	550.68	512	Main	Credit Card	Cancelled	TRK429031	8	FREESHIP	Email				2753.4
5	ORD200003	#####	C31540	Chair	1	273.19	275	Main	Debit Card	Returned	TRK167181	9	WINTER15	Instagram				312.55
6	ORD200004	#####	C81840	Printer	4	626.01	668	Main	Online	Delivered	TRK549301	3	No Coupon	Google				473.96
7	ORD200005	#####	C37249	Phone	2	245.86	934	Main	Credit Card	Shipped	TRK131941	3	FREESHIP	Email				1067.62
8	ORD200006	#####	C83492	Laptop	1	664.42	986	Main	Gift Card	Returned	TRK119721	3	No Coupon	Referral				423.4
9	ORD200007	#####	C41460	Monitor	5	149.55	706	Main	Cash	Shipped	TRK651911	3	SAVE10	Facebook				431.44
10	ORD200008	#####	C26817	Phone	2	134.28	904	Main	Gift Card	Cancelled	TRK591011	9	SAVE10	Google				1120
11	ORD200009	#####	C31946	Desk	4	509.38	102	Main	Credit Card	Shipped	TRK167181	9	WINTER15	Instagram				312.55
12	ORD200010	#####	C43443	Tablet	5	625.97	333	Main	Credit Card	Returned	TRK526461	6	No Coupon	Google				1371.8
13	ORD200011	#####	C93861	Monitor	3	49.14	831	Main	Online	Returned	TRK759371	2	SAVE10	Google				99.24
14	ORD200012	#####	C38785	Monitor	2	180.5	179	Main	Debit Card	Pending	TRK361681	5	FREESHIP	Referral				1093
15	ORD200013	#####	C88348	Laptop	3	201.49	903	Main	Credit Card	Returned	TRK819131	4	FREESHIP	Facebook				604.47
16	ORD200014	#####	C98474	Tablet	2	393.33	980	Main	Debit Card	Pending	TRK791861	3	SAVE10	Instagram				786.66
17	ORD200015	#####	C39416	Printer	1	473.96	942	Main	Cash	Delivered	TRK549301	3	No Coupon	Google				473.96
18	ORD200016	#####	C28976	Printer	2	533.81	914	Main	Credit Card	Pending	TRK131941	3	FREESHIP	Email				1067.62
19	ORD200017	#####	C60672	Tablet	1	423.4	382	Main	Debit Card	Shipped	TRK119721	3	No Coupon	Referral				423.4
20	ORD200018	#####	C84329	Desk	1	431.44	891	Main	Gift Card	Pending	TRK651911	3	SAVE10	Facebook				431.44
21	ORD200019	#####	C14552	Monitor	5	224	273	Main	Cash	Returned	TRK591011	9	SAVE10	Google				1120
22	ORD200020	#####	C54829	Chair	5	62.51	587	Main	Cash	Pending	TRK167181	9	WINTER15	Instagram				312.55
23	ORD200021	#####	C99023	Monitor	4	342.95	569	Main	Cash	Delivered	TRK526461	6	No Coupon	Google				1371.8
24	ORD200022	#####	C34416	Monitor	2	49.62	633	Main	Online	Pending	TRK759371	2	SAVE10	Google				99.24
25	ORD200023	#####	C42372	Phone	4	273.25	755	Main	Debit Card	Cancelled	TRK361681	5	FREESHIP	Referral				1093
26	ORD200024	#####	C17470	Laptop	2	230.95	956	Main	Online	Delivered	TRK287751	3	WINTER15	Referral				461.9
27	ORD200025	#####	C42577	Monitor	1	399.98	709	Main	Debit Card	Delivered	TRK694231	3	WINTER15	Email				399.98
28	ORD200026	#####	C21944	Laptop	5	59.55	296	Main	Online	Delivered	TRK802761	6	No Coupon	Referral				297.75

3. Date Format Validation

Date columns were reviewed to ensure all entries followed a consistent date format.

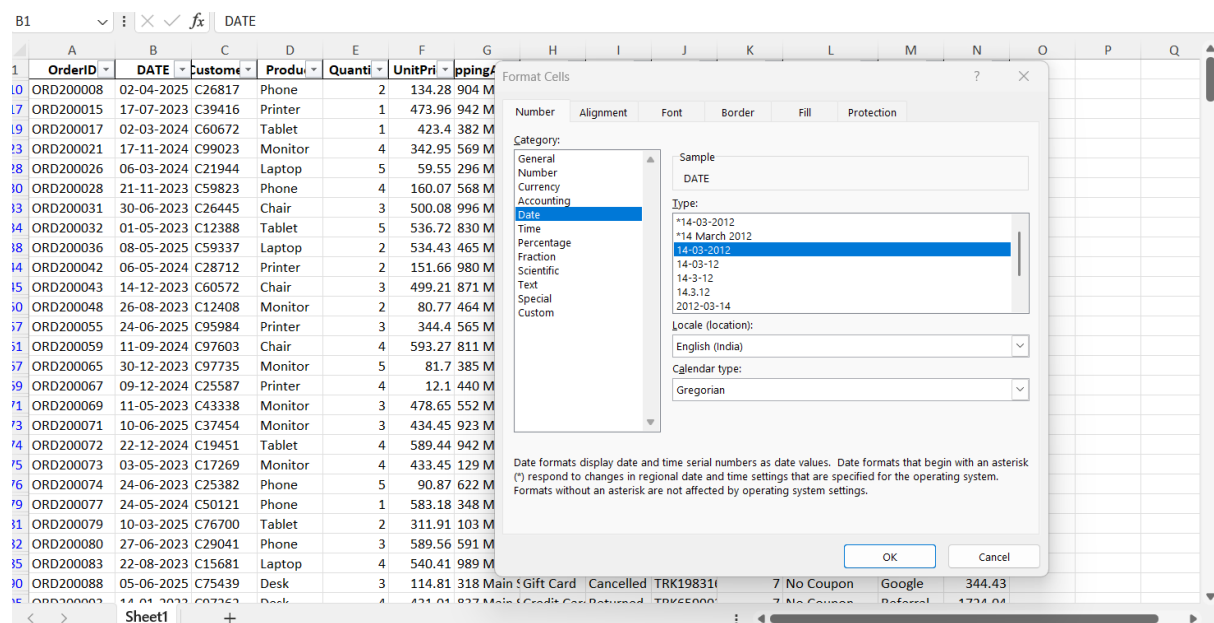
Observation:

- All date values were correctly formatted.
- No invalid date entries were found.

Result:

- Date data was verified and standardized.

Screenshot:



4. Numeric Data Verification

Numeric columns were inspected to ensure they contained valid numerical values and consistent formatting.

Observation:

- Numeric columns were correctly formatted.
- No invalid numeric values were detected.

Result:

- Numeric data integrity was confirmed.

Tools Used

- Microsoft Excel
- Data Filtering
- Conditional Formatting
- Sorting and Validation Techniques

Results Summary

Validation Check	Status
Missing Values Checked	Completed
Missing Values Handled	Completed
Duplicate Records Checked	Completed
Duplicate Order IDs Checked	Completed
Date Formats Verified	Completed
Numeric Columns Verified	Completed

Conclusion

The dataset was successfully cleaned and validated. Missing values were identified and handled appropriately, duplicate records were checked, and all date and numeric fields were verified for consistency. The cleaned dataset is now ready for further analysis and reporting.

This project provided practical experience in data cleaning and preparation, which are essential steps in the data analytics workflow.